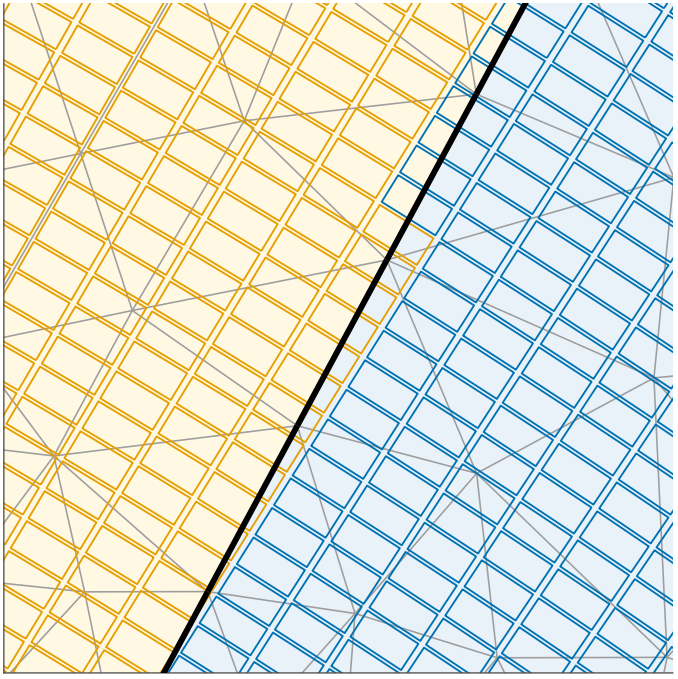
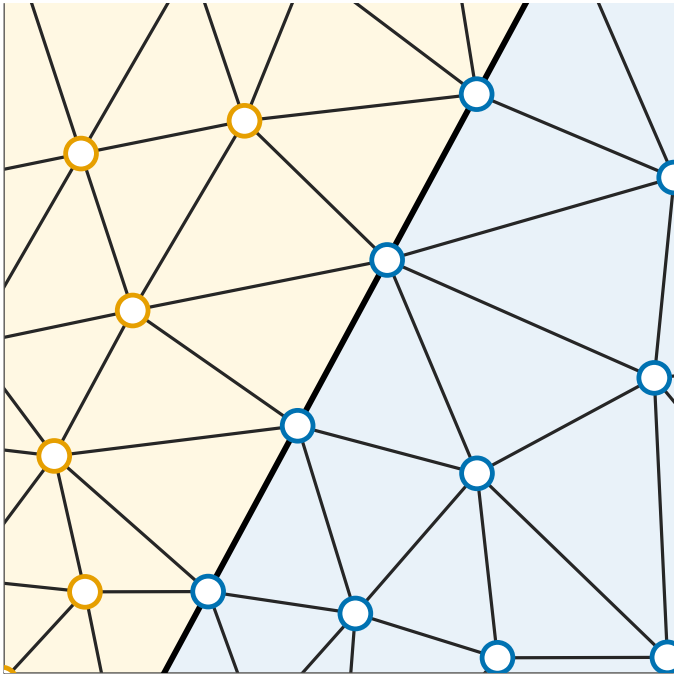
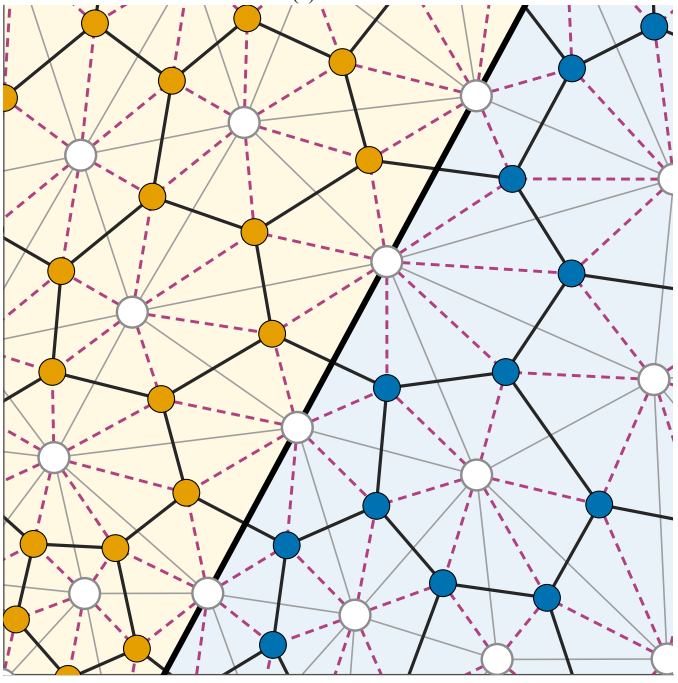


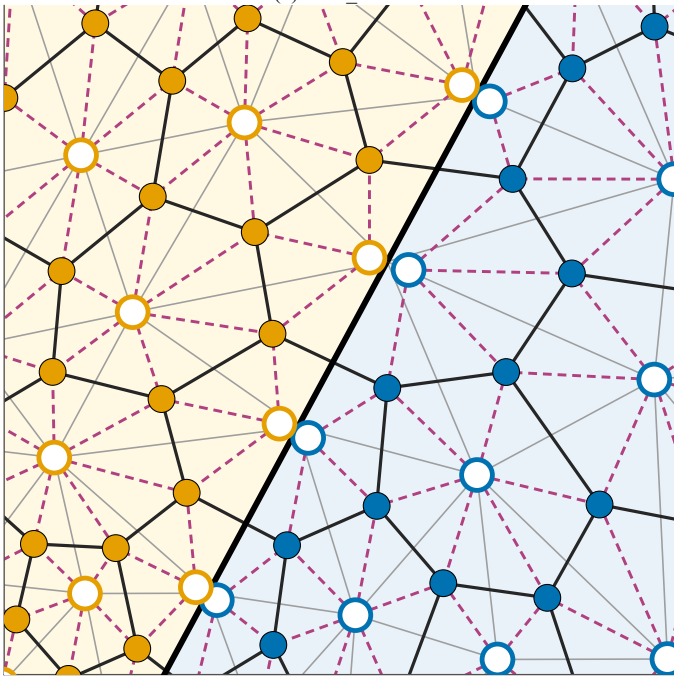
(a) FNO2d



(b) FNO_GNN



(c) FNO_Bipartite



(d) FNO_SplitGNN

Ferro
Cobre
Célula da grade (contorno, cor do material)
Nó de vértice / cópia (anel, cor do material)
Nó de elemento (disco, cor do material)
Vértice sem material (c)
Aresta da malha
Aresta do grafo (vértices, duais, Edge_dual)
Aresta cruzada / Edge_aux
Interface